FX Tracker

Know what your money is really worth

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https://github.com/BElmane/fx-tracker

The Problem

- Exchange rates change every day
- $500 at 9.78 MAD = **4,890 MAD**
- $500 at 9.60 MAD = **4,800 MAD**
- Bad timing = **90 MAD lost**

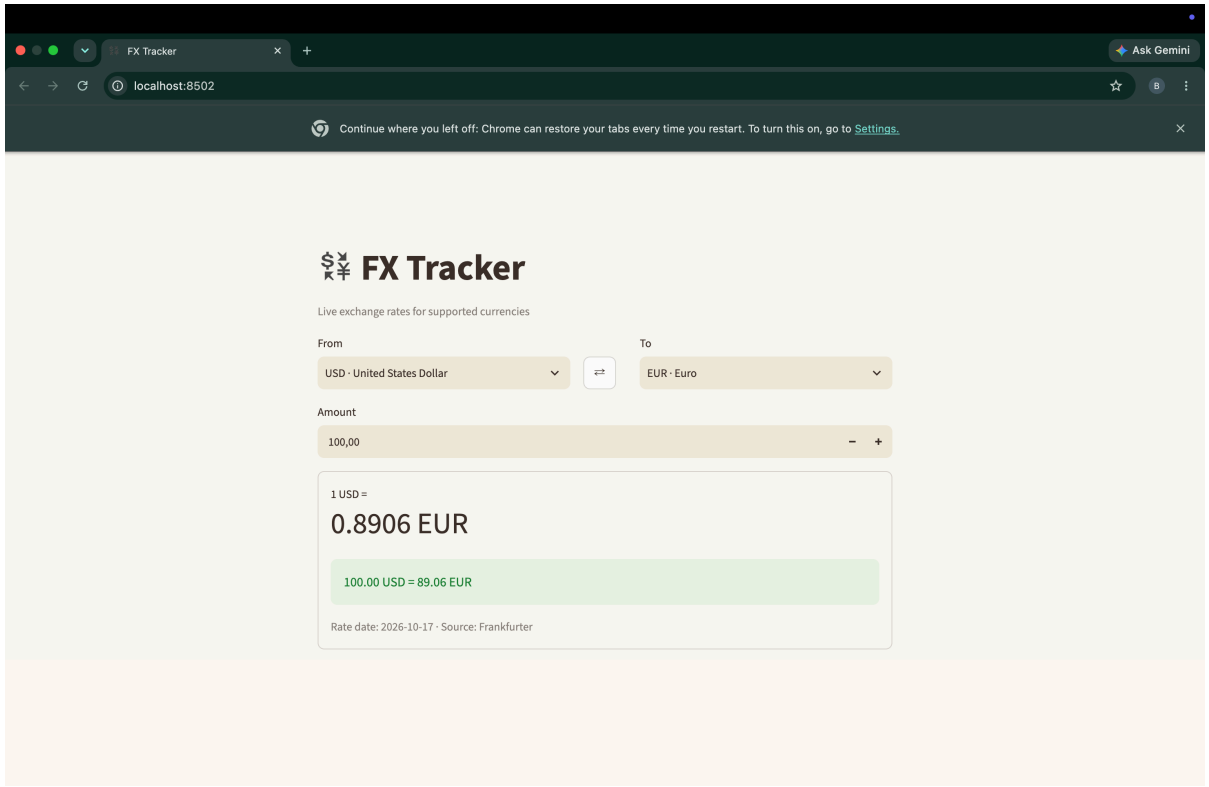
Users: travelers, students whose families send money, anyone who converts money

My Solution

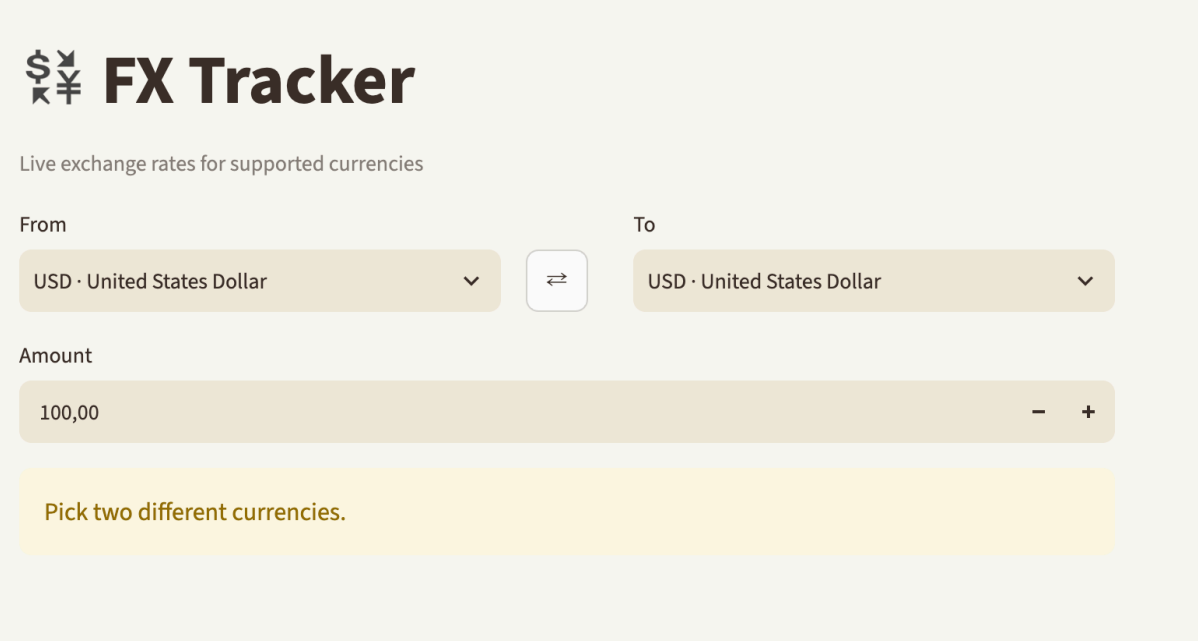
A simple app that shows **live exchange rates**

- 165 currencies
- Real data from the Frankfurter API
- Convert any amount, swap in one click

Demo


FX Tracker
Ask Gemini
localhost:8502
Continue where you left off: Chrome can restore your tabs every time you restart. To turn this on, go to Settings.
FX Tracker
Live exchange rates for supported currencies
From
USD · United States Dollar
To
EUR · Euro
Amount
100,00
1 USD =
0.8906 EUR
100.00 USD = 89.06 EUR
Rate date: 2026-10-17 · Source: Frankfurter

Demo: Errors



Friendly warnings, and the app never crashes

Weekly Progress Sprint 1

- **Week 5:** repo and README, tested the API by hand
- **Week 6:** chose Frankfurter v2 over v1
- **Week 7:** planned the app structure and tests
- **Week 8:** built the full prototype, 35 tests, PR #1 merged

Sprint 1 Metrics

- Lines of code: **258** app, **324** tests
- Tests: **35** (21 unit, 4 integration, 5 regression, 5 acceptance)
- Features: **3 / 3**
- Requirements: **2 / 2**
- Burndown rate for features = **100%**
- Burndown rate for requirements = **100%**

Design

- **Abstraction:** `RateProvider` is a contract for any data source
- **Dependency Injection:** `RateService` gets its provider passed in
- **Result:** tests use a fake provider, no internet needed
- **Refactoring:** moved shared test fakes into one file (`tests/fakes.py`)

What Went Wrong

- Git workflow mistakes, like editing files in two places
- Small mistakes: unsaved files, indentation, a file-name typo
- Learning new tools while building(Streamlit, Apptest)

What Went Well

- The design made testing easy
- All 4 test types done
- The prototype works with real rates

Improvement Plan

- Small commits every week
- Run tests before every commit

Sprint 2 Goals

- Track many currency pairs
- Save rate history and show a trend chart
- Set a target rate and get an alert
- User Manual, Architecture Doc, GitHub Pages

Planned: 5 features, 3 requirements, tests for all 4 types

Sprint 2 Timeline

- **Week 9:** S1P
- **Weeks 10–11:** refactor Sprint 1 code
- **Weeks 12–13:** multi-pair tracking, target rates, docs, website
- **Week 14:** trend view and testing
- **Week 15:** final submission (12/5)

Questions?

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